

Repertoire

FOH CHANNEL EQ SETUP — DiGiCo Quantum 225

Show	Repertoire
Show Date	2026-08-01
Venue	Fountain Square (outdoor)
Console	DiGiCo Quantum 225
Channels	17
FOH Engineer	Brian Lloyd
Monitor Engineer	
Show Time	
Rev	Rev 1.0 Deep Build

DOCUMENT CONTENTS

1. Cover Page — show, venue, and personnel
2. Input List — color-coded full channel patch sheet
3. Patching — AES and Local inputs sorted by port
4. EQ Setup — 17 channels for DiGiCo Quantum 225
5. Reference — EQ structure, pan guide, bus grouping, soundcheck order

SHOW STYLE: A Cincinnati R&B and neo-soul covers set on an open plaza, fronted by an opera-trained lead and backed by a lean band with no horns and no percussion. The vocal carries the show; the two keyboards carry the colour.

Repertoire

VENUE: Fountain Square (outdoor)

DATE: 2026-08-01

REV: Rev 1.0 Deep Build

FOH: Brian Lloyd

MON:

SHOW TIME:

Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
■ DRUMS						
1	Kick In	Shure Beta 91A	Local 1	✓	—	Boundary mic lying inside on the pillow, aimed at the beater. Contour switch assumed FLAT.
2	Kick Out	Shure Beta 52A	Local 2		Short	At the resonant-head port. Two-mic pair with ch 1 — polarity-check the sum in mono at soundcheck.
3	Snare Top	Audix i5	Local 3		Short	Locker fork raised (Lauten LS-408) — Brian kept the i5.
4	Drum Pad	Passive DI (model TBC)	Local 4		DI	Pad model unknown. Patch from the pad's L/MONO output — NOT a Y-cable of L+R. If the DI turns out to be active (J48), it needs 48V.
5	Hat	Shure SM81	Local 5	✓	Boom	Low-frequency filter switch assumed FLAT. e609-style clips don't apply — this needs a boom.
6	Rack 1	Audix D2	Local 6		Clip	Tom sizes unknown — if these are 13"/14", the HPF and the 350 both come down a step.
7	Rack 2	Audix D2	Local 7		Clip	Second of the matched D2 pair from the DP8.
8	Floor	Audix D4	Local 8		Clip	Built on the research reading of the D4, not the old KB row — see mic notes.
9	Overheads	Shure Beta 27 (pair)	Local 9	✓	Boom	STEREO on fader 9 — both mics, one fader. The submitted sheet had OH L/R on 9 and 10; ch 10 is the SNARE PL8 return and is protected. Pad assumed OUT, LF rolloff assumed FLAT. Rain cover — 73–81% chance of rain on a flown condenser pair.
■ RHYTHM						

Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
11	Bass DI	Whirlwind IMP (passive DI)	Local 11		DI	Assumed taken pre-amp, straight off the instrument. Passive DI is –20 dB — the head amp works harder than on an active box. Two-source pair with ch 12.
12	Bass Mic	Shure PG52	Local 12		Short	Brian's own mic (confirmed 2026-07-30). Added to the locker this session — page, mic-library rows and inventory all written.
13	Guitar 1	Electro-Voice N/D 408	Local 13		Short	Mic changed off the sheet's e609 by Brian, 2026-07-30. Amp is an ASSUMPTION — no rider. Bright switch assumed OFF; if it's on, B4 goes to –7. Guitars 2–4 are spares, unused.
17	Key 1	Whirlwind IMP (passive DI)	Local 17		DI	Two separate players, two separate boards. Board assignment is a convention, not a confirmed fact — if Key 2 is the piano, swap the two curves wholesale.
18	Key 2	Whirlwind IMP (passive DI)	Local 18		DI	If either board is a stereo rig summed to one mono DI, confirm it's the board's own mono output and not a Y-cable.

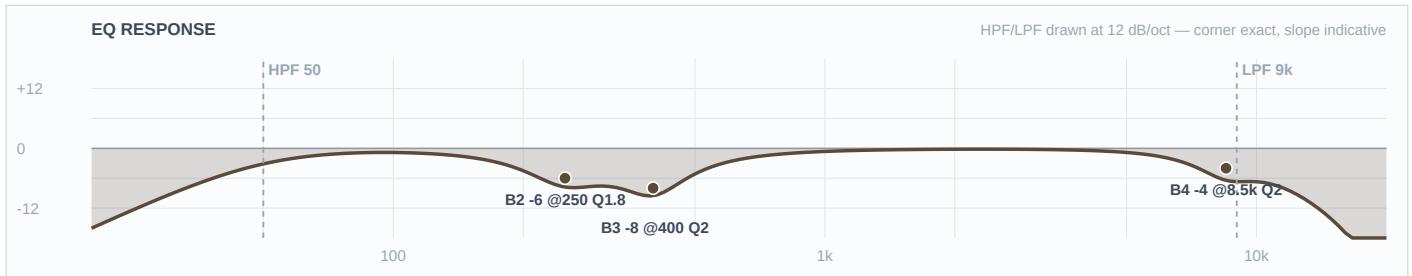
■ VOCALS

33	Brittany	Shure Beta 58A (Wireless 1)	Local 33		—	House wireless 1 → fader 33. This overrides the FSQ template's wireless baseline curve, including its 184 Hz HPF and feedback notch.
34	BG 1	Shure Beta 58A (Wireless 2)	Local 34		—	House wireless 2 → fader 34. Assumed to sing above the lead — if this is a male voice, HPF comes down to ~90 and B2 to ~450.
35	BG 2	Shure Beta 58A (Wireless 3)	Local 35		—	House wireless 3 → fader 35. Wireless 4 (fader 36) unused this show.

■ DRUMS ■

Ch 1 | Kick In

Mic: Shure Beta 91A



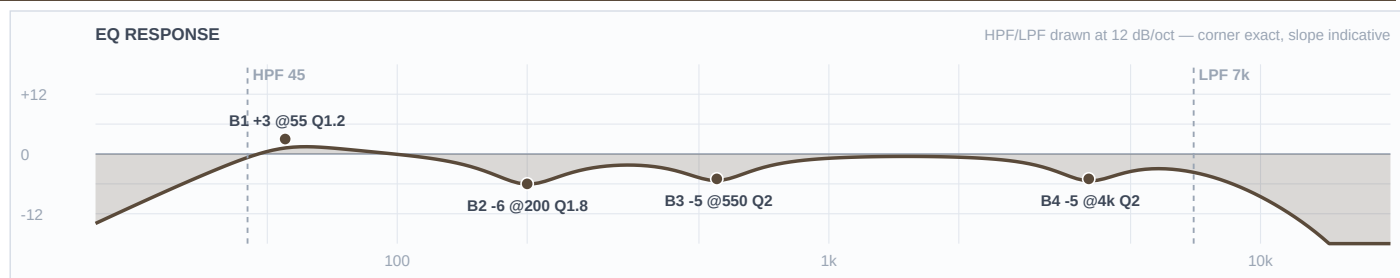
Mic Notes: Shure publish an emphasis from 4 kHz to about 9 kHz for beater attack, and a contour switch that attenuates 400 Hz by 7 dB (Shure USA / RecordingHacks). Both facts drive this channel: nothing gets boosted up top because the capsule already lifts there, and the -8 at 400 is the desk doing by hand exactly what Shure's own switch does — if you find the contour engaged at load-in, halve that cut to -4 or you will do it twice. Boundary mounting also picks up shell boxiness at 300–500 Hz, which is the other half of what B3 and B2 are for.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	50 Hz	LC	—	—	High-pass
4	8.5 kHz	Bell	-4 dB	2	
3	400 Hz	Bell	-8 dB	2	
2	250 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	9 kHz	HC	—	—	Low-pass

Engineer Notes: This is the click channel, and only the click channel. B1 is deliberately FLAT — ch 2 owns the bottom, and a low boost on both kick mics is the classic way to end up with a woolly, phase-smeared kick. The 4–5 kHz part of the capsule's baked emphasis is left completely alone because that IS the beater, and R&B wants it defined rather than absent; the -4 at 8500 takes off the fizz and hat bleed above it, which matters more than usual tonight because 96% humidity means that top end reaches the back of the plaza with almost nothing absorbed. Gate this channel at soundcheck to keep the drum pad's samples out of it.

Ch 2 | Kick Out

Mic: Shure Beta 52A



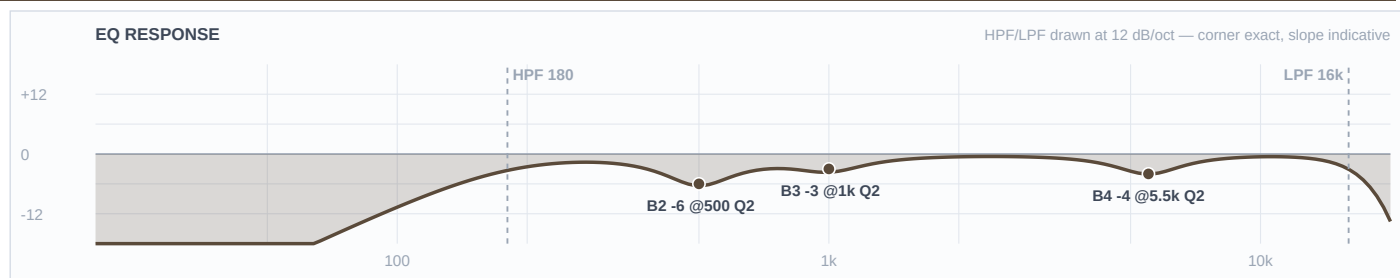
Mic Notes: The 52A is pre-EQ'd from the factory: a big dip around 400 Hz and a peak centred at 4 kHz for beater click, inside a 20 Hz–10 kHz response (Shure Beta 52A spec sheet; Gearspace 'Shure Beta 52 vs Audix D-6'). That 4 kHz peak is real and it is redundant here, because ch 1 already owns the attack — hence the –5 sitting right on it. The 400 Hz dip is why B3 sits BESIDE it at 550 and only goes to –5 instead of the –8 an outdoor show would normally take: cutting hard into a scoop the capsule already provides hollows the drum out.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	45 Hz	LC	—	—	High-pass
4	4 kHz	Bell	-5 dB	2	
3	550 Hz	Bell	-5 dB	2	
2	200 Hz	Bell	-6 dB	1.8	
1	55 Hz	Bell	+3 dB	1.2	
HC	7 kHz	HC	—	—	Low-pass

Engineer Notes: The weight. This is the one channel on the show that earns a low boost, and +3 at 55 is deliberately modest — eight KS21s per side in a delayed arch already own everything under 45 Hz, so this is shaping the kick's body, not reaching for sub. It is also the first thing to come out if the subs read hot at soundcheck. LPF at 7000 keeps this mic entirely out of the attack lane so the two kick channels never fight. If the drum pad turns out to be carrying 808s, pull this boost to FLAT and let the pad be the sub.

Ch 3 | Snare Top

Mic: Audix i5



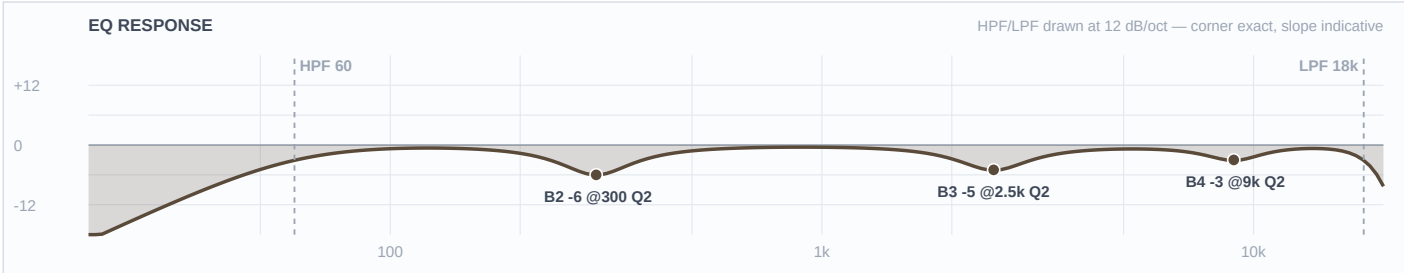
Mic Notes: The i5 bakes in +9 dB at 5500 Hz plus a body lift around 150 Hz, with mids slightly scooped against an SM57 (RecordingHacks / Sound On Sound). Those are the two things a snare EQ normally reaches for, and this mic supplies both — so there is no crack boost and no body boost on this channel at all. B1 stays FLAT because the 150 Hz lift is already the fat.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	180 Hz	LC	—	—	High-pass
4	5.5 kHz	Bell	-4 dB	2	
3	1 kHz	Bell	-3 dB	2	
2	500 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The backbeat of an R&B set, which means fat and controlled rather than cracking. The -4 at 5500 lands squarely on the capsule's peak and runs deeper than it would on a dry night, because at 96% humidity the air absorbs almost nothing and that peak arrives across the plaza intact. The box cut sits at 500 rather than the reflex 300–400: neo-soul's low-mid warmth is the sound of the genre, and pulling 300 Hz out of a snare in this band would cost more than the mud is worth. B3 -3 at 1000 takes the papery ping off so the drum reads round. Gate with a generous range — ghost notes have to live.

Ch 4 | Drum Pad

Mic: Passive DI (model TBC)



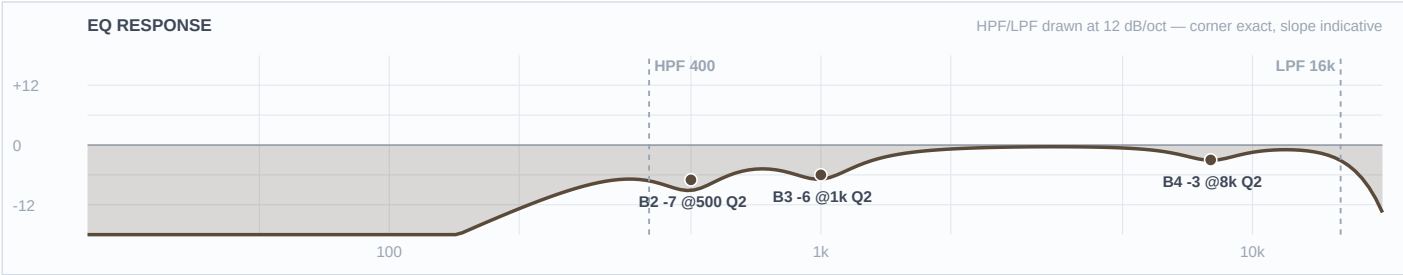
Mic Notes: No capsule — this is a line-level, already-produced stereo bus arriving through a DI. The class-standard unit for this role (Roland SPD-SX) carries its own master EQ and 21 master effects and outputs MASTER OUT L/MONO + R (Roland), so whatever the desk does here sits on top of a mix the drummer already made. There is no KB row for a sampling pad at all — same gap flagged on the 2026-07-27 build — so this unit is verdicted THIN and built conservatively.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-3 dB	2	
3	2.5 kHz	Bell	-5 dB	2	
2	300 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	18 kHz	HC	—	—	Low-pass

Engineer Notes: The one genuine unknown left on this show is whether this pad carries 808s, and that decides who owns 50–80 Hz. HPF at 60 is the hedge: a real 808 fundamental still speaks from 60 up, but the pad is not handed the same octave the kick already has. Two one-line corrections at load-in — if it IS 808-heavy, drop this HPF to 30 and pull ch 2's +3 at 55 to FLAT; if it is claps and FX only, raise this to 120 and leave the kick alone. Nothing is boosted because the samples were mastered by somebody else; the -3 at 9000 is purely the weather, since saturated air delivers pre-hyped sample tops with no absorption at all.

Ch 5 | Hat

Mic: Shure SM81



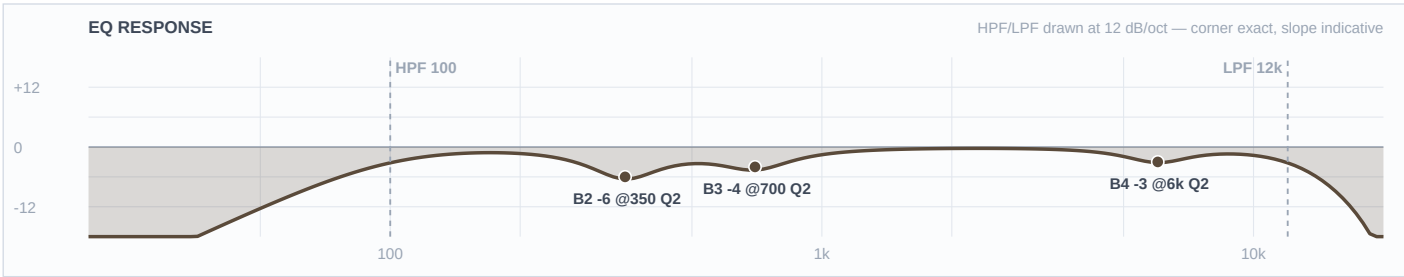
Mic Notes: Ruler-flat 20 Hz–20 kHz with a switchable low-frequency filter at 6 or 18 dB per octave (Shure USA / RecordingHacks). Nothing baked in, nothing to ease off — which also means nothing flatters this source, so every move here is deliberate. If you find the 18 dB/oct filter engaged, drop the desk HPF to 150 so the two don't compound into a thin, papery hat.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	400 Hz	LC	—	—	High-pass
4	8 kHz	Bell	-3 dB	2	
3	1 kHz	Bell	-6 dB	2	
2	500 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The hat carries the subdivision in R&B — the 16ths are what make the groove read as this genre — but with a stereo overhead pair open on ch 9 this is a spot mic, not the cymbal picture. HPF at 400 is high on purpose: gusts to 28 mph, and everything below that is kick and snare bleed. The move worth explaining is the one that isn't here — standard hat guidance is an 8–10 kHz shimmer boost, and tonight that is actively wrong. At 96% humidity the cymbal's real top arrives at the back of the plaza unattenuated, so the correct move is the -3 at 8000, not a lift.

Ch 6 | Rack 1

Mic: Audix D2



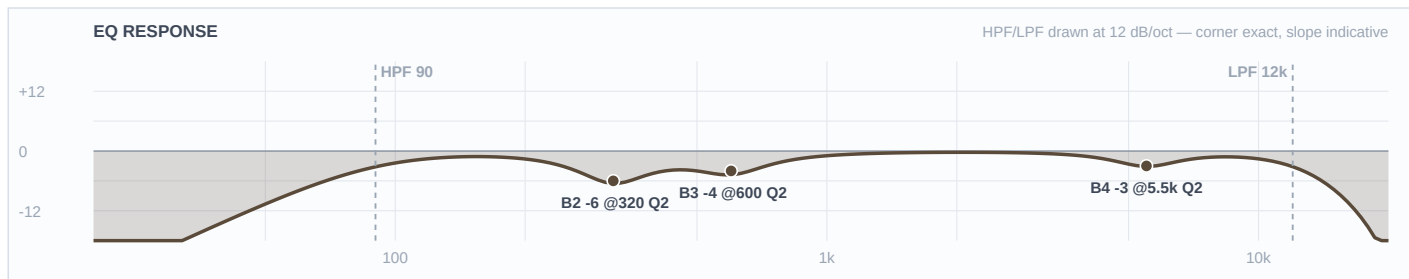
Mic Notes: The D2 boosts around 150 Hz, dips between 500 Hz and 1 kHz, and adds a subtle upper-mid lift, over 68 Hz–18 kHz (RecordingHacks, reading Audix's chart). B1 is FLAT because that 150 Hz lift IS the tom body — a low boost here would stack straight onto it. B3 runs -4 instead of the outdoor -6 for the same reason in reverse: the capsule already scoops 500 Hz–1 kHz, and cutting hard into a baked scoop empties the drum out.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	100 Hz	LC	—	—	High-pass
4	6 kHz	Bell	-3 dB	2	
3	700 Hz	Bell	-4 dB	2	
2	350 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: R&B tom fills want warmth and speed, not attack — they land and they're gone. Nothing on this channel reaches for stick click; the -3 at 6000 actually takes the capsule's upper-mid lift back down, which matters at 96% humidity when that region arrives intact. This channel and Rack 2 share no value anywhere: 6000/700/350 against 5500/600/320, HPF 100 against 90. That separation is what keeps two toms from reading as one blurred thing in an open plaza with no room to blend them.

Ch 7 | Rack 2

Mic: Audix D2



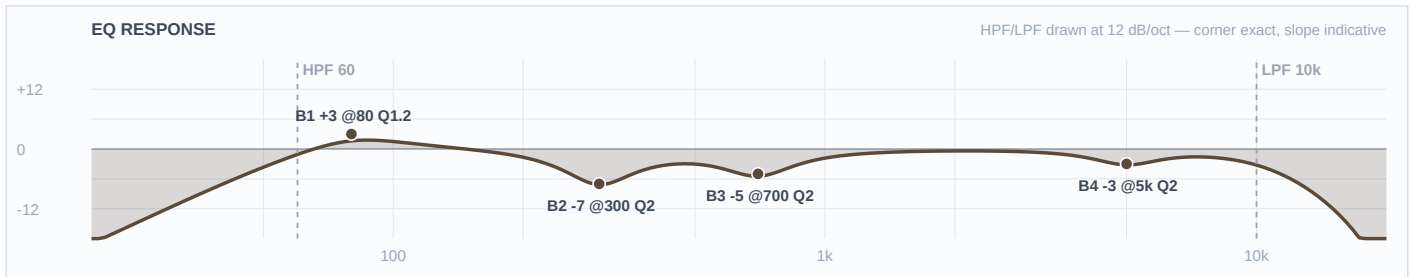
Mic Notes: Same capsule as ch 6 — +150 Hz body, 500 Hz–1 kHz dip, subtle upper-mid lift (RecordingHacks / Audix). Same two consequences: B1 FLAT so nothing stacks on the baked body lift, and B3 held to -4 rather than outdoor depth because the capsule already dips there.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	90 Hz	LC	—	—	High-pass
4	5.5 kHz	Bell	-3 dB	2	
3	600 Hz	Bell	-4 dB	2	
2	320 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: The lower of the two rack toms, and every band sits below its partner deliberately — 5500 against 6000, 600 against 700, 320 against 350, HPF 90 against 100. Slotting the pair this way means a fill descending across them reads as movement instead of as the same drum twice. Warmth over attack, same as ch 6, because that's what the genre asks of a tom.

Ch 8 | Floor

Mic: Audix D4



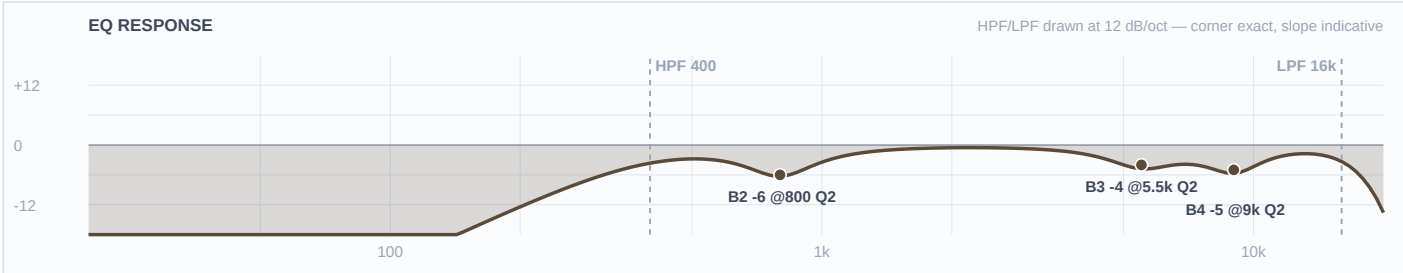
Mic Notes: This is the one web-vs-KB conflict on the show and Brian settled it. Audix's current published chart, read by RecordingHacks, shows +6 dB at 5 kHz with a secondary peak above 10 kHz and a rolloff below 70 Hz. The old mic-library row said the opposite at both ends — 'reaches 35 Hz' and 'less upper-mid attack, may need a touch of click.' Brian's call: go with the research AND fix the KB row. That flips two moves: no click boost at 5 kHz (it would have landed on a +6 dB baked peak), and a +3 at 80 to restore weight the rolloff costs.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	5 kHz	Bell	-3 dB	2	
3	700 Hz	Bell	-5 dB	2	
2	300 Hz	Bell	-7 dB	2	
1	80 Hz	Bell	+3 dB	1.2	
HC	10 kHz	HC	—	—	Low-pass

Engineer Notes: Weight under the fills, which is all an R&B floor tom is asked for. The +3 at 80 is the only low boost outside the kick, and it exists because two things line up: the capsule rolls off below 70, and an open plaza gives back nothing — indoors the room would supply this for free. The -3 at 5000 is the move that would have gone the other way under the old KB row, and it's a useful reminder of why the capsule gate exists. Box cut at -7 runs full outdoor depth here, because on a drum shell that really is mechanical mud, not genre warmth.

Ch 9 | Overheads

Mic: Shure Beta 27 (pair)



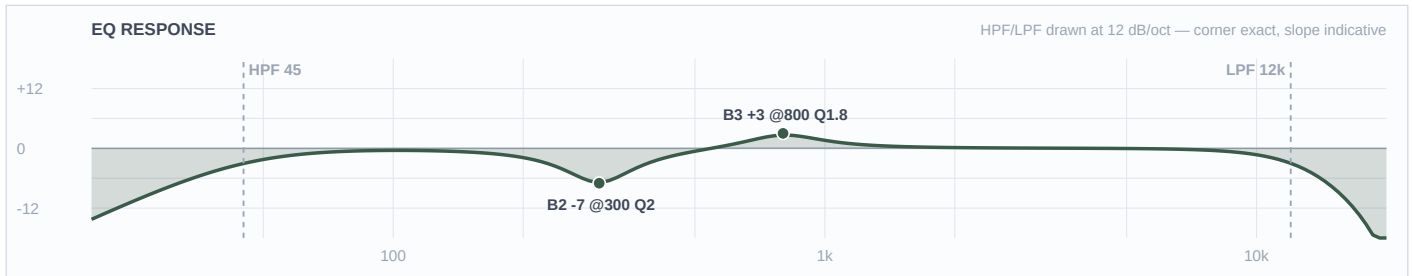
Mic Notes: Nominally flat from about 60 Hz to 3 kHz with two small HF peaks of +2 dB or less at 5500 Hz and 9 kHz, and a supercardioid pattern that stays consistent below 6400 Hz without widening toward omni (RecordingHacks). Both HF bands on this channel sit exactly on those two peaks — the two frequencies an overhead EQ normally reaches for are the two this capsule already supplies, so the desk subtracts. If the LF rolloff switch is engaged at load-in, drop the desk HPF to 150 so they don't compound.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	400 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-5 dB	2	
3	5.5 kHz	Bell	-4 dB	2	
2	800 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The most weather-driven channel on the show. HPF at 400 is a third higher than the 300 that worked five days ago at this venue, and the reason is simply that these gusts are double — 22 to 28 mph through the first three hours, on the pair of mics standing highest and most exposed on the stage. In R&B these are wash and glue, never a drum-kit picture: the close mics do the work and the cymbals sit behind the vocal. Both baked peaks come down because at 96% humidity they arrive across the plaza with no air absorption to soften them, and -6 at 800 handles cymbal honk and snare-bleed body. Worth deciding early how these get covered if the rain comes.

Ch 11 | Bass DI

Mic: Whirlwind IMP (passive DI)



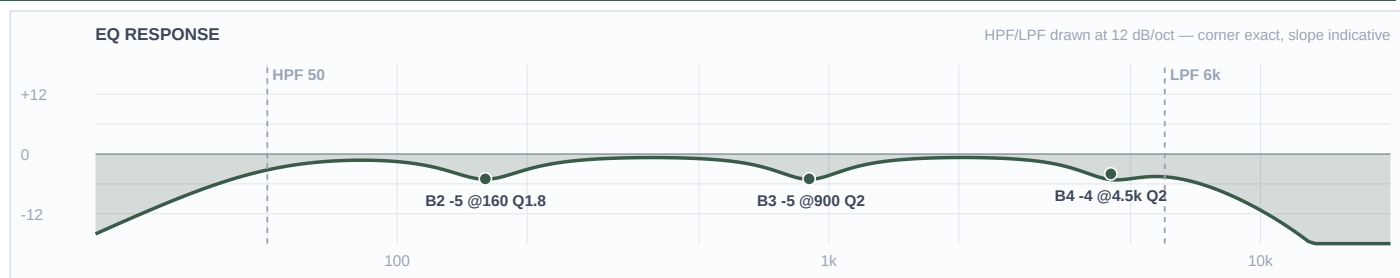
Mic Notes: 20 Hz–20 kHz within ± 1 dB, 133:1 ratio, -20 dB insertion loss, TRHL transformer (Whirlwind spec via Sweetwater / Full Compass). Nothing is baked in anywhere, which is why the one boost on this channel is uncontroversial — there is no capsule feature at 800 Hz to stack onto. Lane split with ch 12 is driven by the PG52's measured curve, not by convention: the cab mic owns 60–100 Hz thump, this channel owns note definition at 800 and everything above.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	45 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	800 Hz	Bell	+3 dB	1.8	
2	300 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: After the vocal this is the most important channel in an R&B band — the pocket between the kick and the bass is the genre. The +3 at 800 is where the note actually speaks, and it is the one lane this channel owns outright, which is why ch 12 is trimmed at 900 rather than left to fight for it. B1 is FLAT on purpose: the DI's fundamental passes through unboosted so it doesn't stack on the cab mic's baked 80 Hz hump. The -7 at 300 is the one place the full outdoor depth applies to the bass, because that is where the DI and the cab genuinely pile onto each other. If it turns out to be a post-amp send, the amp's voicing is already on it and the 800 boost comes out.

Ch 12 | Bass Mic

Mic: Shure PG52



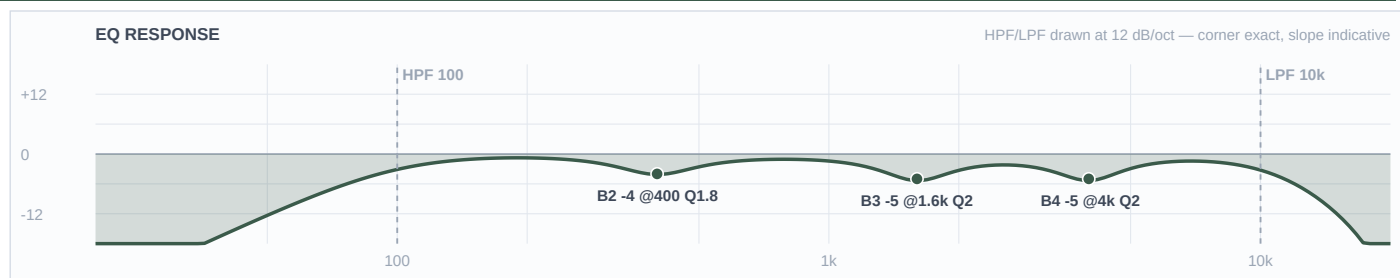
Mic Notes: Shure's own 2008 spec sheet gives 30 Hz–13 kHz, 300 Ω , -55 dBV/Pa, neodymium, internal shock mount, and a printed curve showing a low hump around 60–100 Hz, a broad dip through roughly 200–800 Hz, a presence rise at 4–5 kHz and a hard rolloff above 10 kHz. Independently, the forums put it a class below a Beta 52 on kick but name bass cabinets as the one place it excels — 'you can get fantastic results with it' — and describe exactly the same 80 Hz hump (Gearspace; HomeRecording; TalkBass). Whoever specced this put the mic on the source its own users rate it for.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	50 Hz	LC	—	—	High-pass
4	4.5 kHz	Bell	-4 dB	2	
3	900 Hz	Bell	-5 dB	2	
2	160 Hz	Bell	-5 dB	1.8	
1	—	OFF	—	—	Flat
HC	6 kHz	HC	—	—	Low-pass

Engineer Notes: Three of these four bands were placed by the capsule rather than by taste. B1 is FLAT because the mic already humps at 60–100 — this channel owns the cab-thump lane by letting that hump through, not by boosting it. B2 sits at 160 rather than the reflex 250 because 250 is inside the capsule's own broad dip and cutting there would hollow the cab out. B4 trims the baked 4–5 kHz rise instead of adding to it, since ch 11 owns definition anyway. And -5 at 900 is pure lane discipline: 800 is the DI's boost, so the cab mic steps just above it. LPF at 6000 keeps cab hiss and stage bleed off an open plaza.

Ch 13 | Guitar 1

Mic: Electro-Voice N/D 408



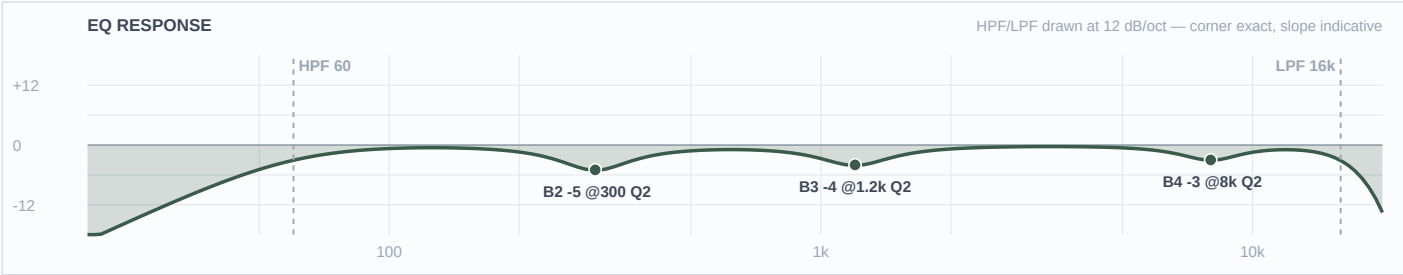
Mic Notes: Two documented facts stack here and they both point the same way. The N/D 408 is a supercardioid N/DYM dynamic that is brighter and more aggressive through the upper mids than an SM57, with a deliberate proximity lift (EV N/D 408B data sheet, 1992; corroborated by EV's own product copy listing it for guitar amp). The Fender blackface Twin tone stack dips from about 80 Hz to 500 Hz and rises from roughly 1,300 Hz upward (Gearspace; fenderguru.com). So an upper-mid-forward capsule is pointed at an amp that is already top-forward. Worth noting the swap improved this: the e609's own 4–5 kHz presence lift would have been the third bright thing in the chain.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	100 Hz	LC	—	—	High-pass
4	4 kHz	Bell	-5 dB	2	
3	1.6 kHz	Bell	-5 dB	2	
2	400 Hz	Bell	-4 dB	1.8	
1	—	OFF	—	—	Flat
HC	10 kHz	HC	—	—	Low-pass

Engineer Notes: A neo-soul comping guitar is a texture, and its job here is to stay out of Brittany's way — which is what the -5 at 1600 is for, carved a full octave below where the vocals are cut (4000/3500/4500) so the two are never fighting the same band. The -5 at 4000 is the deepest high-band cut on the show, and it takes three things to earn that: an aggressive capsule, a bright amp, and saturated air that absorbs almost nothing across the throw. B1 is where the equipment layer wanted a boost and the capsule overruled it — the Twin scoops 80–500 and would normally earn a +3 at 200, but at 2–3 inches off the grille the 408's proximity lift already piles low-mids in exactly that region. If the guitar reads thin at soundcheck, move the mic, not the EQ.

Ch 17 | Key 1

Mic: Whirlwind IMP (passive DI)



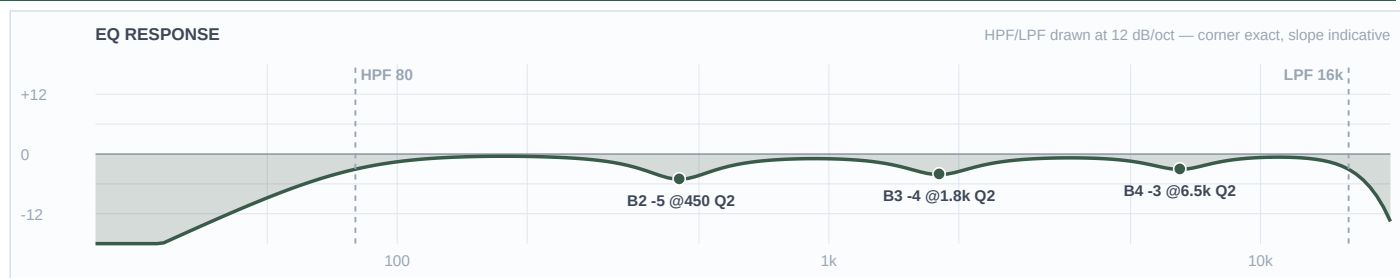
Mic Notes: Passive DI, 20 Hz–20 kHz ±1 dB, nothing baked in (Whirlwind spec) — so every move on this channel is a separation decision rather than a correction. There is no KB row for a keyboard or line-level board at all, the same gap logged on the 2026-07-27 build, so this unit is verdicted THIN and the values are conservative.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	8 kHz	Bell	-3 dB	2	
3	1.2 kHz	Bell	-4 dB	2	
2	300 Hz	Bell	-5 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: With no horns and no percussion in this band, the two keyboards carry all the harmonic colour, so keeping them off each other and off the vocal is the whole job. The low-mid cut runs -5, not the -8 the outdoor rule would give, and that is a deliberate trade: the FSQ deep-cut rule exists to fight mud, and outdoors that mud is mechanical — box, bleed, cab resonance — not room buildup, because an open plaza has none. A Rhodes' 250 Hz warmth is not mud, it is what the genre is made of. The depth the venue rule wants gets spent on the drums and the guitar cab instead. B3 at 1200 clears the vocal band from underneath.

Ch 18 | Key 2

Mic: Whirlwind IMP (passive DI)



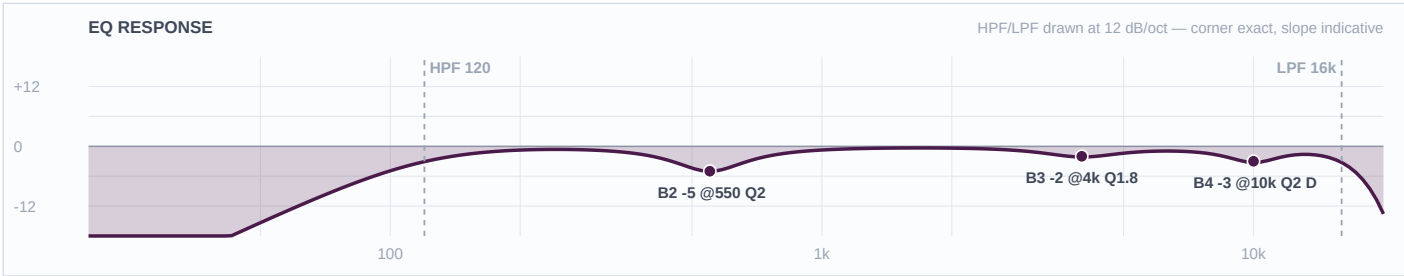
Mic Notes: Same passive IMP, same ± 1 dB flat response, same absence of anything to compensate for (Whirlwind spec). The separation from ch 17 is entirely deliberate rather than capsule-driven.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	80 Hz	LC	—	—	High-pass
4	6.5 kHz	Bell	-3 dB	2	
3	1.8 kHz	Bell	-4 dB	2	
2	450 Hz	Bell	-5 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The second board sits clear of the first in every band — HPF 80 against 60, 6500 against 8000, 1800 against 1200, 450 against 300. No shared frequency, no shared depth. In a band with two keys and nothing else filling the middle, that slotting is what stops the two players from turning into one indistinct pad. Both B3 carves sit around the vocals' cuts rather than on them, so the keys clear Brittany's band from underneath. Same genre-over-venue call as ch 17 on the low-mids: -5, not -8.

Ch 33 | Brittany

Mic: Shure Beta 58A (Wireless 1)



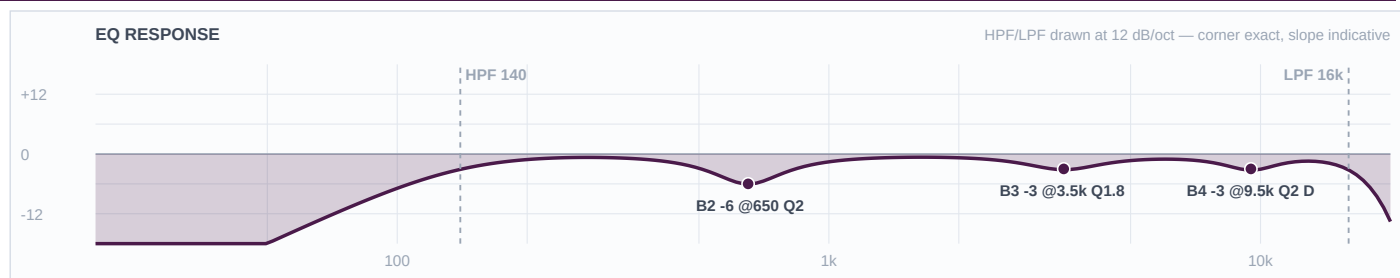
Mic Notes: The Beta 58A carries two presence peaks, at roughly 4 kHz and 10 kHz, over 50 Hz–16 kHz, with a gradually falling bass response below 500 Hz for proximity control (Shure USA; Music On Stage technical analysis 2026). Those peaks are what makes it cut and also what makes it strident — both come down here rather than being left. Cuts only, as every vocal channel gets.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	120 Hz	LC	—	—	High-pass
4	10 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	4 kHz	Bell	-2 dB	1.8	
2	550 Hz	Bell	-5 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: Brittany Marie is the show. She entered Cincinnati's SCPA at eight and trained in jazz, classical and contemporary plus two years with the Cincinnati Opera, and that is not trivia — it means a supported, projecting instrument with real dynamic range that will back off the mic on the big notes. Two decisions come straight from it. The template's 184 Hz HPF is wrong for her and goes to 120: a female R&B lead sings down to around F3–G3, so 184 would sit on top of her lowest fundamentals — the same class of error caught on a male bass voice at this venue in July. And the 4 kHz cut stays light at -2 because she does not need help cutting through this band; anything more and you lose the warmth the genre is built on. The de-esser is dynamic because humidity climbs from 93% to 99% across the show and a static top-end value set at six o'clock will be wrong by eleven.

Ch 34 | BG 1

Mic: Shure Beta 58A (Wireless 2)



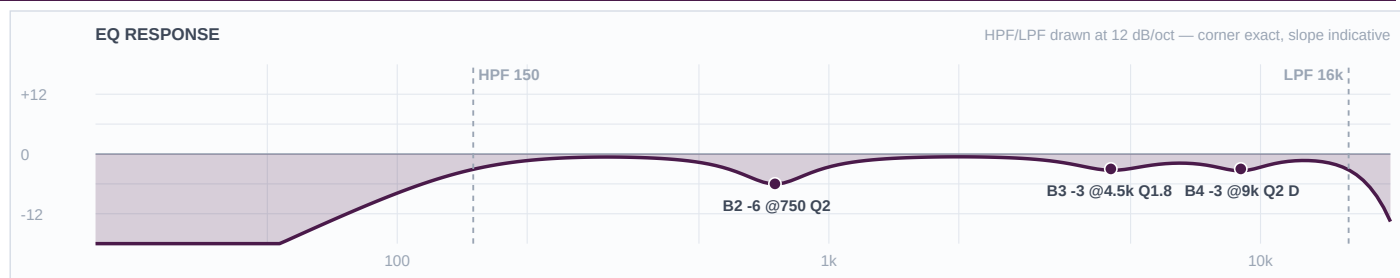
Mic Notes: Same capsule as ch 33 — baked peaks near 4 kHz and 10 kHz, sub-500 Hz proximity roll (Shure USA; Music On Stage). Every band here is a trim of something the mic already supplies. Cuts only.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	140 Hz	LC	—	—	High-pass
4	9.5 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	3.5 kHz	Bell	-3 dB	1.8	
2	650 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: Sits under Brittany by design rather than by fader alone. This channel is cut deeper than hers in the presence region (-3 at 3500 against her -2 at 4000) and deeper in the box (-6 against -5), so she comes out on top without her channel ever being boosted — which matters, because vocal boosts are the thing that costs you the show outdoors. HPF at 140 is higher than the lead's: a backing vocal has less to lose below there and more to gain in feedback margin with three open mics and gusts on the stage.

Ch 35 | BG 2

Mic: Shure Beta 58A (Wireless 3)



Mic Notes: Third Beta 58A on the house wireless — same two baked presence peaks, same proximity roll (Shure USA; Music On Stage). Cuts only.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	150 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	4.5 kHz	Bell	-3 dB	1.8	
2	750 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: The second backing voice is slotted clear of the first as well as of the lead: 9000/4500/750 against BG 1's 9500/3500/650, and HPF 150 against 140. No two of the three vocal channels share a single value in any band, which is what keeps a three-part stack sounding like three people instead of one thick smear across an open plaza. Deeper presence cut than the lead, same as BG 1, so Brittany sits above both.